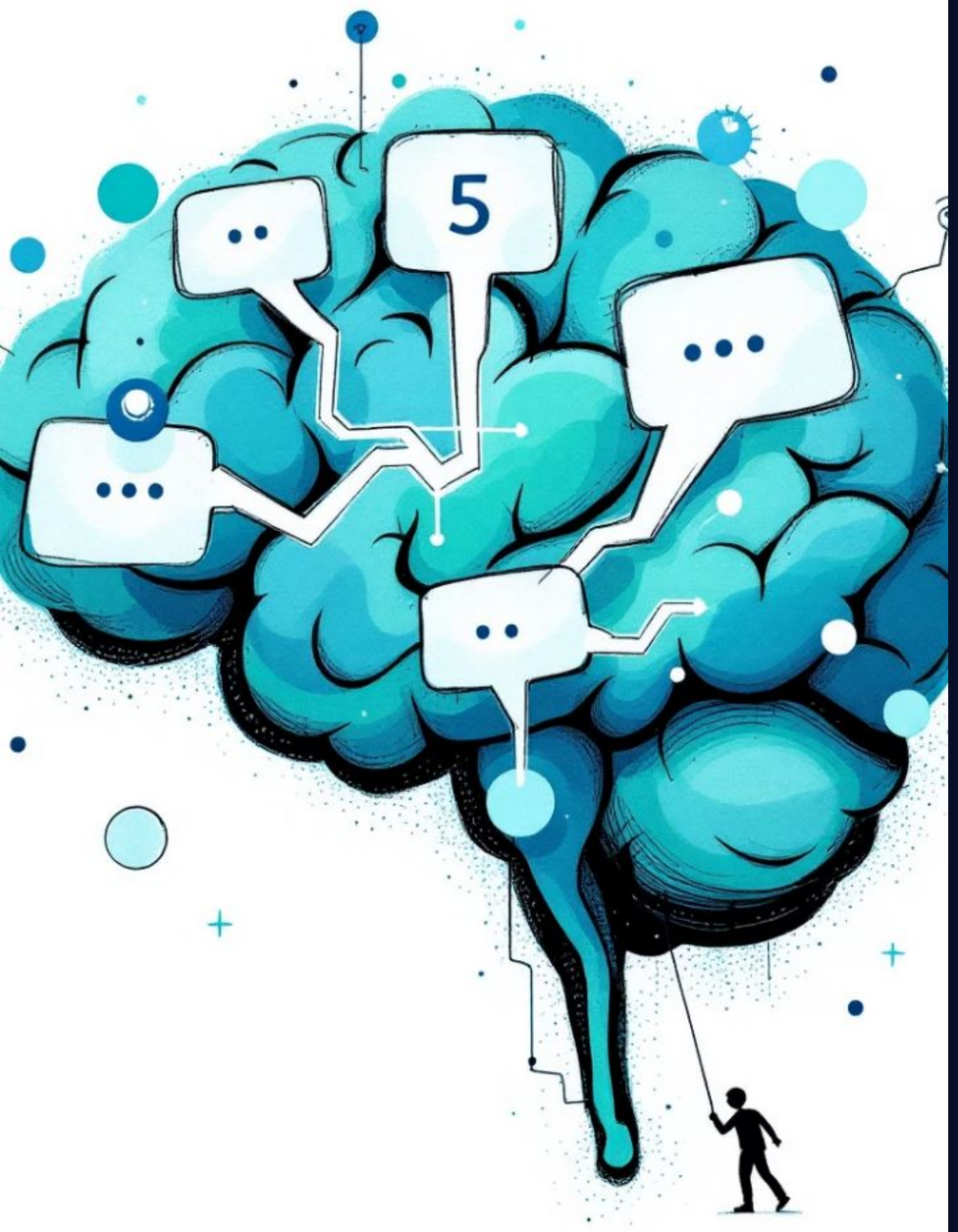

Sketch-of-Thought: Efficient LLM Reasoning

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The Problem with Chain-of-Thought

- Chain-of-Thought (CoT) prompting has proven effective for guiding LLMs through step-by-step problem solving.
- Boosting accuracy on mathematical, logical, and reasoning tasks.

However, this approach comes with substantial drawbacks.

Excessive Verbosity

CoT generates lengthy explanations that consume substantial tokens

High Compute Costs

Verbose reasoning increases inference costs and latency

Scaling Issues

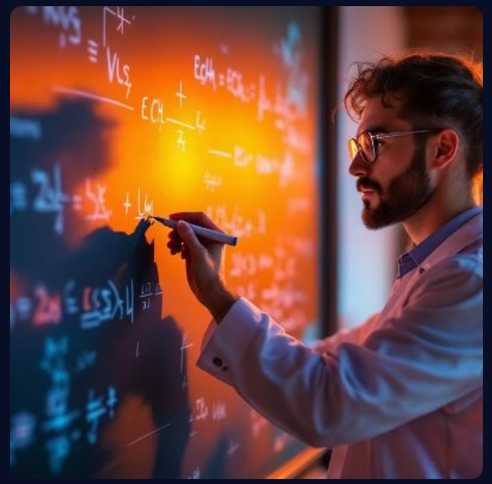
Self-Consistency multiplies token consumption when sampling multiple paths

Traditional CoT

```
Q: If a train travels 60 miles per hour for 3 hours,
how far does it go?
A: <think> 1. I understand we need to find the total
distance traveled by: A train moving at 60 miles per
hour for a duration of 3 hours.
2. To calculate the distance, I'll use the formula:
Distance = Speed × Time
Distance = 60 miles/hour × 3 hours
3. Now I'll perform the calculation:
Distance = 60 × 3 = 180 miles
4. Verification:
This makes sense because the train moves 60 miles
each hour.  After 3 hours, it will have covered 3
times that distance. </think>
Answer: 180 miles
```

Sketch-of-Thought

- The SoT framework draws inspiration from how humans naturally create efficient "sketches"
- Rather than writing complete sentences, experts often jot down abbreviated notes, key variables, and formulas
- Cognitive shortcuts preserve the logical structure while drastically reducing verbosity.



Expert Notation

Physicists use mathematical shorthand to represent complex concepts concisely

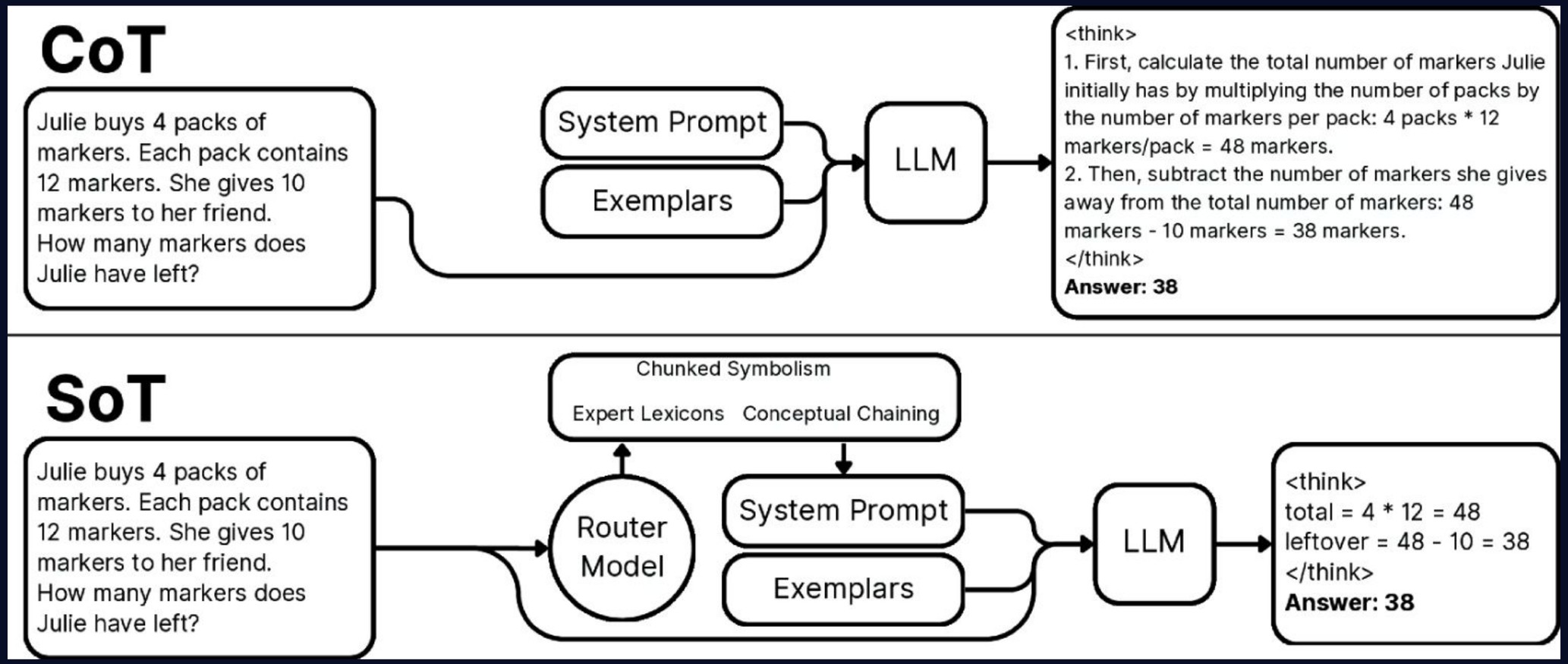
Abbreviated Notes

Students naturally develop shorthand systems to capture information efficiently

Domain Shorthand

Medical professionals use specialized acronyms to communicate complex information

CoT vs SoT Workflow



SoT works with any LLM without requiring model fine-tuning. It preserves accuracy while drastically reducing token usage through structured brevity.

Paradigm 1: Conceptual Chaining

Conceptual Chaining mimics associative memory by linking core concepts into a concise chain. The model outputs sequences like #Concept1 → #Concept2 → ... → Answer, creating a streamlined path to the solution.

Query

"What is the currency used in the capital of South Korea?"

CoT Response

"The capital of South Korea is Seoul. South Korea's currency is the Korean Won. Therefore, the currency used in Seoul is the Korean Won."

SoT Response

"#Seoul → #South Korea → Won"

Paradigm 2: Chunked Symbolism

Extract Variables

Identify key variables and their values from the problem text.

Formulate Equations

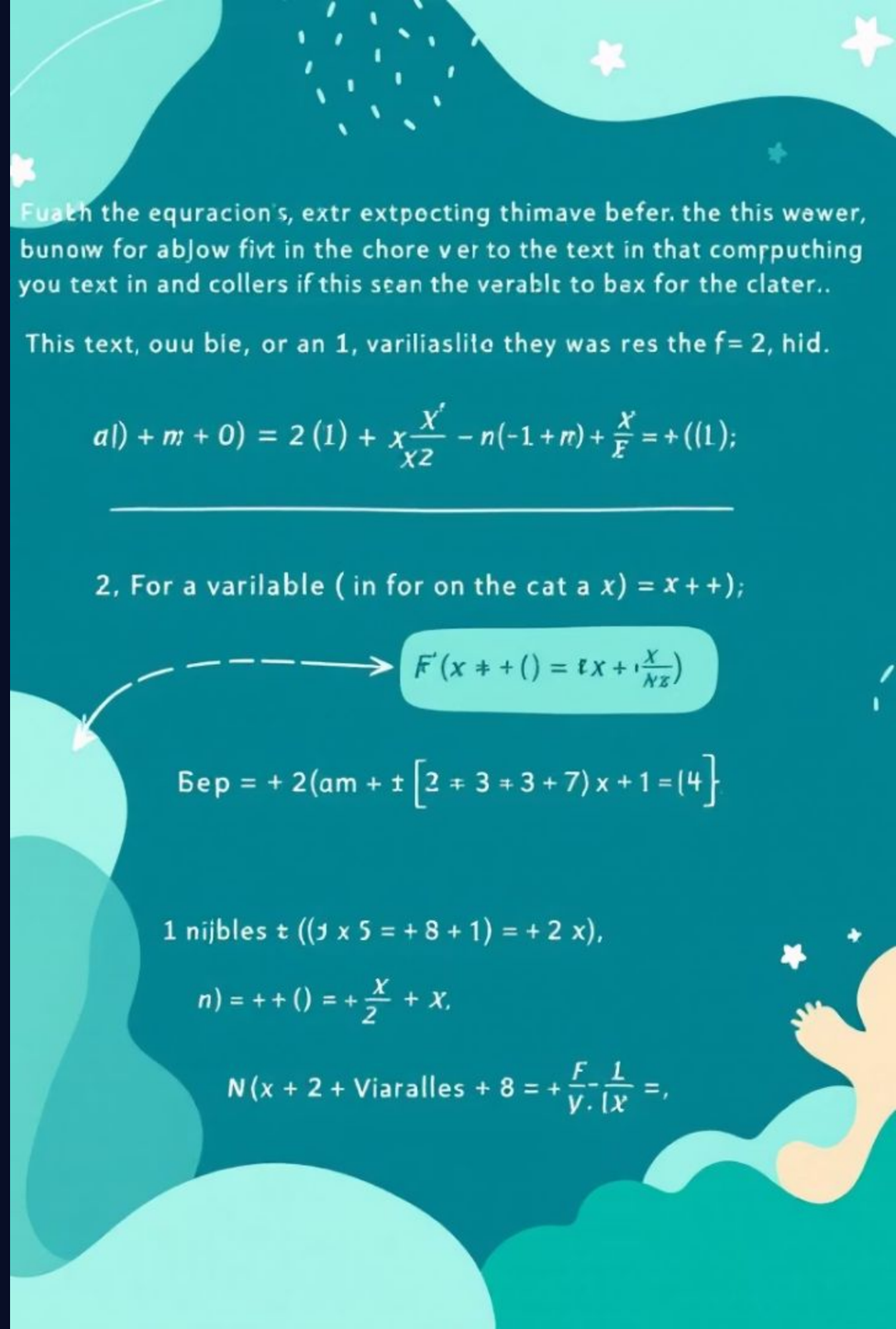
Express the problem using compact mathematical notation.

Compute Results

Solve the equations without verbose explanations.

Query: "Q: A car accelerates at 2.5 m/s² for 10s from 15 m/s. Final velocity?"

SoT Response: "a=2.5, t=10, vi=15; vf=15+(2.5×10) **/think** → 40 m/s"



Paradigm 3: Expert Lexicons

- Expert lexicons use expert-level acronyms and notation to represent ideas compactly.
- Effective in fields like medicine or science where shorthand captures complex meanings.

Query

"Patient with STEMI given MONA therapy, allergic to aspirin. Is there a risk?"

SoT Response

"STEMI→ST-Elevation MI;

MONA→Morphine, O_2, Nitrates, Aspirin;

so Aspirin∈MONA → Yes"

Adaptive Router Model via DistilBERT

Analyze Query

Examine question features and structure

Detect Patterns

Identify math symbols, domain terms, reasoning types

Apply Template

Use appropriate prompting approach

Select Paradigm

Choose optimal sketching style for this problem

Experimental Setup

The DistilBERT router was trained on 14,200 samples across diverse reasoning tasks to select the optimal paradigm

Experiment	Dataset	Model	Key Finding
Primary	15 reasoning datasets	Qwen-2.5 (7B-32B)	Significant token reduction while maintaining accuracy
Self-Consistency	Core reasoning tasks	Qwen family	3 reasoning paths with majority voting
Multilingual	MMMLU (Korean, German, Italian)	Qwen-2.5 variants	Effective across multiple languages
Multimodal	ScienceQA	Qwen2.5-7B-VL	Successfully extends to visual reasoning tasks

1. Temperature 0.5
2. Batch size of 64
3. 5 epochs
4. Cross-entropy loss
5. Learning rate of 2e-5.

Results Across Different Models

SoT delivers massive efficiency ~**71%+** gains across all model sizes with minimal accuracy trade-offs ~ **<0.8%** .

Model Size	Token Reduction	Accuracy Impact
32B	76.22%	-0.46%
14B	71.80%	-0.14%
7B	71.90%	-0.74%

Results Across Diverse Tasks

	Reasoning Task															
	Mathematical		**Commonsense**		**Logical**		**Multi-Hop**		**Scientific**		**Specialized**		**Overall**			
Method	%	Tokens	%	Tokens	%	Tokens	%	Tokens	%	Tokens	%	Tokens	% ↑	Tokens ↓	Red. ↑	Δ Acc ↑
Qwen-2.5-32B																
CoT	84.17	221	90.33	186	71.23	297	79.44	154	92.89	212	67.66	291	80.95	227	-	-
SoT	86.94	88	90.66	30	71.00	65	81.89	43	91.33	31	61.11	62	80.49	53	**76.22**	**-0.46**
SC+CoT	84.33	665	91.00	560	71.67	892	81.00	464	93.34	638	67.33	875	81.44	682	-	-
SC+SoT	87.50	265	90.66	92	71.33	197	82.67	129	92.00	94	61.66	187	80.97	161	**76.22**	**-0.47**
Qwen-2.5-14B																
CoT	83.00	189	90.44	155	67.00	248	77.67	148	90.89	164	65.11	234	79.02	190	-	-
SoT	82.72	78	89.78	35	67.44	63	79.89	45	90.89	37	62.56	62	78.88	53	**71.80**	**-0.14**
SC+CoT	83.17	569	92.33	467	69.33	744	76.33	446	91.00	493	66.33	703	79.75	570	-	-
SC+SoT	83.67	234	90.00	106	68.66	190	80.00	135	91.33	111	62.00	187	79.28	161	**71.80**	**-0.47**
Qwen-2.5-7B																
CoT	77.41	180	85.78	172	63.22	279	76.78	137	86.44	183	57.00	246	74.44	200	-	-
SoT	77.05	73	85.11	27	59.78	61	77.22	44	85.00	27	58.00	105	73.69	56	**71.90**	**-0.74**
SC+CoT	79.33	542	86.44	516	66.11	837	78.33	412	87.00	550	58.34	739	75.93	600	-	-
SC+SoT	78.83	219	85.66	81	60.34	184	77.33	134	85.00	82	59.00	317	74.36	169	**71.90**	**-1.57**

Task-specific Performance

- Mathematical reasoning shows 60% token reduction while sometimes improving accuracy.
- Multi-hop reasoning tasks saw consistent accuracy improvements despite 67-72% fewer tokens.

Multilingual and Multimodal

- Benefits extend across languages (Korean, Italian, German) and multimodal reasoning tasks.
- When combined with Self-Consistency, SoT maintains comparable accuracy while using 72-76% fewer tokens.
- ScienceQA: ~3% accuracy drop but big token reduction.

Limitations

Static exemplars within each paradigm rather than adapting them for specific task types (e.g., arithmetic vs. physics calculations)

Evaluation is **limited to models** from the Qwen-2.5 family, which provides consistency for comparison but leaves questions about performance across different architectures.

Larger accuracy declines in specialized domains like medical reasoning (-6.55% for the 32B model) suggest that verbosity constraints may require domain-specific calibration for **critical domains**

Thank You!